

Computer Vision

Milestone 1 — Literature Review & Exploratory Data Analysis

Due Date	Weighting	Submission
Tuesday 11:59 PM	5% of Final Grade	LMS (ZIP archive)

Overview

In Milestone 1, students establish the analytical foundation for their Computer Vision project. This involves surveying relevant prior work through a structured literature review and gaining a thorough quantitative and visual understanding of the dataset that will drive the project. Strong completion of this milestone directly informs sound model design decisions in later milestones.

Learning Outcomes

- Critically evaluate and synthesize CV research papers
- Identify key datasets, benchmarks, and state-of-the-art methods
- Profile a dataset statistically (class balance, image sizes, quality)
- Produce meaningful EDA visualisations (histograms, sample grids, heatmaps)
- Communicate findings clearly in a technical written report

Part A — Literature Review

Students must survey a minimum of 10 peer-reviewed papers (conference or journal) published within the last five years that are directly relevant to their chosen CV task (e.g., object detection, segmentation, image classification, depth estimation).

Required Sections

Section	Description	Min. Length
Problem Context	Define the CV task, its real-world relevance, and open challenges.	~200 words
Paper Summaries	For each paper: objective, methodology, dataset used, key metric results, and limitations.	~150 words each
Comparative Table	Side-by-side comparison of methods across accuracy, speed, and dataset used.	1 table
Gap Analysis	Identify what your project addresses that existing work does not.	~150 words

References	IEEE or APA format. All citations must appear in-text.	Full list
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Recommended Sources

- IEEE CVPR, ICCV, ECCV — top CV conference proceedings
- arXiv.org (cs.CV) — preprints; prefer peer-reviewed where possible
- Google Scholar, Semantic Scholar — for citation tracking and related work
- Papers With Code — benchmarks and reproducible results

Part B — Exploratory Data Analysis (EDA)

Using the dataset selected for the project, students must perform a systematic EDA and document findings in a Jupyter Notebook submitted alongside the written report. All figures must include titles, axis labels, and a brief caption explaining what the figure shows.

Required Analyses

#	Analysis	What to Include
1	Dataset Summary	Total images, number of classes, train/val/test split sizes, data source and license.
2	Class Distribution	Bar chart or pie chart of samples per class. Comment on any imbalance and its implications.
3	Image Statistics	Histogram of image dimensions. Report mean, median, min/max pixel values across the dataset.
4	Sample Visualisation	5x5 grid of random samples per class with labels. Identify any noise or annotation errors.
5	Colour & Texture Analysis	Channel-wise pixel intensity distributions (R/G/B or grayscale). Optional: average image per class.
6	Correlation / Heatmaps	For structured metadata (bounding box sizes, aspect ratios), provide correlation matrices or scatter plots.
7	Data Quality Notes	Document missing files, corrupt images, duplicate samples, or label inconsistencies found.

Submission Instructions

Upload a single ZIP archive named TeamID_M1.zip containing:

- Literature Review as a PDF
- EDA Jupyter Notebook (.ipynb with outputs)
- README.txt with dataset download instructions

Academic Integrity

All written content must be original. Direct quotations from papers must be cited and limited to under 10% of the report.

- Disclose use of AI writing tools in an appendix
- Code must be individually written — sharing notebooks across teams is considered plagiarism
- All cited papers must be accessible via provided links or DOIs

